Literature Survey: Time-Series Foundation Models on Indian Data

Binay Verma · B.Tech CSE (Data Science), RV University · October 2026

1. Scope and research question

This survey reviews published work relevant to one question: **how well do pretrained time-series foundation models (TSFMs) forecast Indian time series, whose structure includes lunar-calendar festivals and region-specific calendar effects, compared with classical models given India-specific calendar inputs?** It covers four areas: the main TSFMs, the benchmarks used to evaluate them and their known weaknesses, research on calendar and holiday inputs to TSFMs, and existing work on festival effects and TSFMs using Indian data. The survey is based on web and arXiv searches carried out on 27 October 2026.

2. Time-series foundation models

A TSFM is a large model pretrained once on many time series so that it can forecast a new series *zero-shot*, i.e. without being trained on that series. Table 1 lists the models most often used as reference points in the literature reviewed here.

Model	Developer, venue	Core idea	Inputs
Chronos [1]	Amazon, TMLR 2024	Scales and quantises values into a fixed token vocabulary and trains T5-style language-model architectures on the tokens; evaluated on 42 datasets.	Univariate
TimesFM [2]	Google, ICML 2024	Decoder-only transformer with input patching, about 200M parameters, pretrained on about 100B time points, mostly Google Trends search-interest series and Wikipedia pageviews.	Univariate (later versions add covariates [14])
Moirai [3]	Salesforce, ICML 2024	Masked-encoder transformer with any-variate attention and multiple patch sizes; trained on LOTSA, an open archive of 27B+ observations across nine domains.	Multivariate
Chronos-2 [4]	Amazon, arXiv 2025	Group attention enables in-context learning across related series, variates and covariates; reported state-of-the-art results on fev-bench, GIFT-Eval and Chronos Benchmark II.	Univariate, multivariate, past-only and known-future covariates

Table 1. Main TSFMs referenced in this survey.

Chronos-2's technical report gives a directly relevant example: on Rossmann store-sales data, the univariate forecast was nearly flat, while the version given promotion and holiday covariates captured the actual sales pattern [4].

3. Benchmarks for TSFMs

Benchmark	Composition	Finding relevant to this survey
GIFT-Eval [5]	28 datasets, over 144,000 series, seven domains; includes a non-leaking pretraining set.	Original Moirai models scored better on datasets that leaked from their training data, and the effect grew with prediction length.
fev-bench [6]	100 tasks across seven domains, 46 with covariates; bootstrapped confidence intervals.	Existing TSFMs often lose accuracy because they ignore covariates.
TIME [7]	50 fresh datasets, 98 tasks, built for leakage-free zero-shot evaluation; 12 TSFMs evaluated (ICML 2026).	Argues existing benchmarks reuse legacy sources and use task setups detached from real operational needs.

Benchmark	Composition	Finding relevant to this survey
IND-Eval [8]	Industrial time series organised into 187 source-specific tasks.	TimesFM-3 led on seven of nine sources; the exceptions are used to argue for domain-representative evaluation.
TempusBench [9]	New datasets not in TSFM pretraining corpora (NeurIPS 2025 workshop).	Notes that existing benchmarks overlook non-stationarity and seasonality, and tune baselines unfairly.

Table 2. Benchmarks. None of those found is built around Indian data.

4. Known weaknesses of TSFMs

4.1 Data leakage and contamination

Because pretraining corpora are large and often not fully disclosed, test data may already have been seen in pretraining, which inflates zero-shot scores [5]. Meyer et al. add a second form, *temporal* leakage: a model trained on one series (e.g. bus ridership) can gain information about a correlated test series from the same period (e.g. metro ridership) [11]. Pan and Ezzat, evaluating TSFMs for electricity price forecasting, therefore added a newly curated 2025 US dataset whose evaluation period postdates the disclosed pretraining data of Chronos-2 and TimesFM 2.5 [12].

4.2 Dependence on pretraining domains

Karaouli et al. report that a TSFM's zero-shot ability is strongly tied to the domains it was pretrained on, and that fine-tuned TSFMs did not consistently beat smaller task-specific models by a margin justifying their size [13].

4.3 LLM-based forecasters

Tan et al. (NeurIPS 2024) ablated three methods that repurpose large language models for forecasting; removing the language model or replacing it with a basic attention layer did not hurt accuracy, and often improved it [10]. This critique concerns repurposed text LLMs, not models pretrained on time series such as Chronos or TimesFM.

5. Calendar and holiday inputs to TSFMs

Kreusel et al. (KIT, 2026) is the closest methodological precedent [14]. They compared seven ways of encoding hour, weekday and holiday information across four TSFMs (Chronos-2, TabPFN-TS-3, TiRex-2, TimesFM-3) on nine hourly datasets with one test year each. Key points:

- Encoding choice changed MASE by up to 9.9% on non-holidays and up to 83% on holidays within a single model.
- Gains from calendar inputs were concentrated on holidays, which cannot be inferred from the series' own history.
- Holidays were added as flags from the Python *holidays* library; the datasets come from Madrid, New York, Washington DC, Alicante and the German state of Baden-Württemberg.
- The authors note a limitation: the context of 8,192 hourly steps is about 49 weeks, so a full annual cycle never fits in the model's input.

Hertel et al. used SHAP explanations on Chronos-2 and TabPFN-TS for German load data and showed the holiday feature lowering the forecast on public holidays [15].

6. Moving festival effects in Indian time series

Indian festivals such as Diwali and Navaratri follow the lunar calendar, so their date in the Gregorian calendar changes every year. Classical (non-TSFM) work has already measured this:

- **Bhattacharya, Pandey, Patnaik and Shah (NIPFP Working Paper 160, 2016)** [16] seasonally adjusted IIP, exports, CPI and WPI with X-13ARIMA-SEATS. Diwali, which falls in October or November, had a significant effect on IIP; Eid did not. Adding a Diwali regressor reduced the standard deviation of the adjusted month-on-month IIP growth from 22.74 to 21.23; for October

2009, the measured growth changed from −28.6% to −12.98%.

- **Sanyal and Mitra (RBI), with McElroy and Roy (US Census Bureau), 2017** [17] studied Navaratri and Diwali in 13 RBI series. For IIP consumer durables, the estimated festival windows began 12 days before Navaratri and 10 days before Diwali. Only two of the thirteen series both needed seasonal adjustment and showed a significant holiday effect. They also note that festival dates and celebrations differ by region, so national aggregates smooth out regional effects.

Both studies use RegARIMA models (seasonal ARIMA with regression inputs) with festival-window regressors. This is the established classical method for these effects.

7. TSFMs applied to Indian data

- **Koley et al. (Journal of Banking and Financial Technology, 2026)** [18] evaluated several TSFMs against non-foundation models on Indian card-industry metrics from RBI data, in a data-limited setting, and reported performance advantages for TSFMs. (Only the abstract was available; the full paper is paywalled.)
- **FedChronos (IIT Ropar, 2026)** [19] fine-tuned Chronos-T5-Tiny on daily commodity prices from 15 Indian agricultural markets in nine states. Naive LoRA fine-tuning overfit the small per-market datasets and performed worse than the zero-shot model.
- **Indian datasets.** A Karnataka SLDC load dataset (116 days in 2024) states that no public benchmark exists for Indian state-level SLDC data [20]. A Delhi hourly load and weather dataset covering January 2018 to March 2025 is available on IEEE DataPort [21].

8. Research gap

Based on the sources found, the following gaps exist:

- No multi-domain benchmark evaluates TSFMs specifically on Indian time series (Section 3).
- Studies of calendar inputs to TSFMs use Western datasets and holiday calendars; lunar-calendar festivals such as Diwali and Navaratri have not been tested (Section 5).
- Indian festival effects are documented with classical RegARIMA models, but those models have not been compared against TSFMs on the same series (Sections 6 and 7).
- Existing TSFM studies on Indian data cover single domains and do not evaluate festival periods separately (Section 7).

Caveat: this survey is search-based and may have missed relevant work, especially very recent preprints. The gap should be re-checked before writing a paper.

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